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Machine Learning in Practice

Supervised, Unsupervised & Probabilistic Algorithms for Engineering

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Preface & Curriculum Overview

This textbook has been compiled for undergraduate and postgraduate students enrolled in the **AI & Machine Learning (CSM)** program. It provides structured, rigorous theoretical models, practical case demonstrations, analytical formulas, and chapter-end review exercises. Students are encouraged to practice the core derivations and apply algorithmic workflows to real-world engineering challenges.

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Chapter 1: Foundations of Statistical Learning

1.1 Supervised vs Unsupervised Learning

Statistical learning models seek functions mapping input vectors to output target spaces by minimizing empirical risk over sample distributions.

> ***Key Concept:*** *The bias-variance tradeoff determines model generalization on unseen evaluation distributions.*

1.2 Gradient Descent Optimization

First-order optimization methods iteratively update weight vectors proportionally to the negative gradient of the objective cost function.

Review Questions & Laboratory Exercises

- Derive the closed-form Ordinary Least Squares estimator for multi-variable regression.
- Explain why L1 regularization produces sparse feature representations.

Chapter 2: Classification and Decision Boundaries

2.1 Logistic Regression & Cross-Entropy

Binary and multinomial classification utilize sigmoid and softmax activations with negative log-likelihood loss formulations.

> ***Key Concept:*** *Maximum likelihood estimation under Bernoulli distributed targets yields binary cross-entropy.*

2.2 Support Vector Machines & Kernels

Maximum-margin hyperplanes maximize geometric margins while mapping non-linear inputs into high-dimensional Hilbert spaces via Mercer kernels.

Review Questions & Laboratory Exercises

- State and prove the Karush-Kuhn-Tucker (KKT) dual formulation for soft-margin SVMs.
- Compute the confusion matrix metrics: Precision, Recall, and F1-score.

Chapter 3: Ensemble Methods & Model Evaluation

3.1 Random Forests & Bagging

Bootstrap aggregating reduces estimator variance by training de-correlated decision trees across stochastic feature sub-spaces.

> ***Key Concept:*** *Boosting constructs sequential additive models focusing residuals on misclassified samples.*

Review Questions & Laboratory Exercises

- Compare the gradient boosting algorithm with AdaBoost.
- Explain k-fold cross validation protocols.